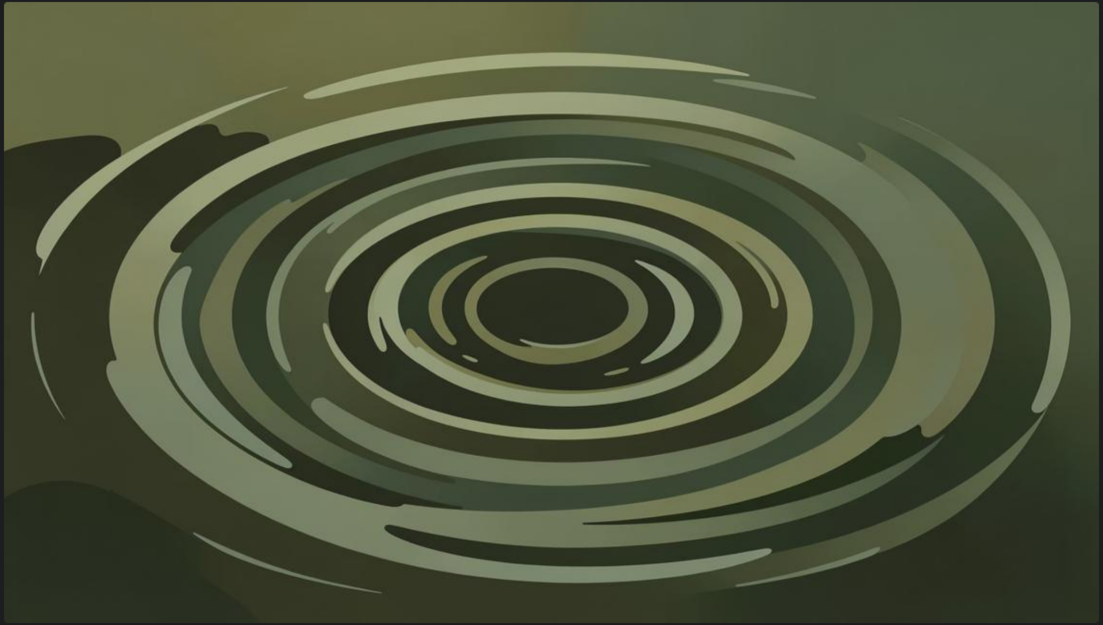
Group 2: Infinite Impulse Response (IIR) Filters

A deep dive into recursive filter design, stability, phase behavior, and classical design methodologies.

DIGITAL SIGNAL PROCESSING



What is an IIR Filter?

Unlike FIR filters, IIR filters have an impulse response that **theoretically lasts forever** — hence the name. This "infinite echo" stems from their recursive nature: the output depends on both past inputs *and* past outputs.

This feedback mechanism allows IIR filters to achieve sharp frequency responses with far fewer coefficients than FIR filters — making them significantly more computationally efficient.

Recursive Structures: The Feedback Feedback Loop

IIR filters are also called **recursive digital filters** because of their inherent feedback mechanisms. The governing difference equation is:

$$y[n] = \sum_{k=0}^{M} b_k \ u[n-k] - \sum_{k=1}^{N} a_k \ y[n-k]$$

Feedback Coefficients (a_k)

Past outputs y[n−k] feed back into the current output — the defining feature of IIR filters.

Feedforward Coefficients (b_k) (b_k)

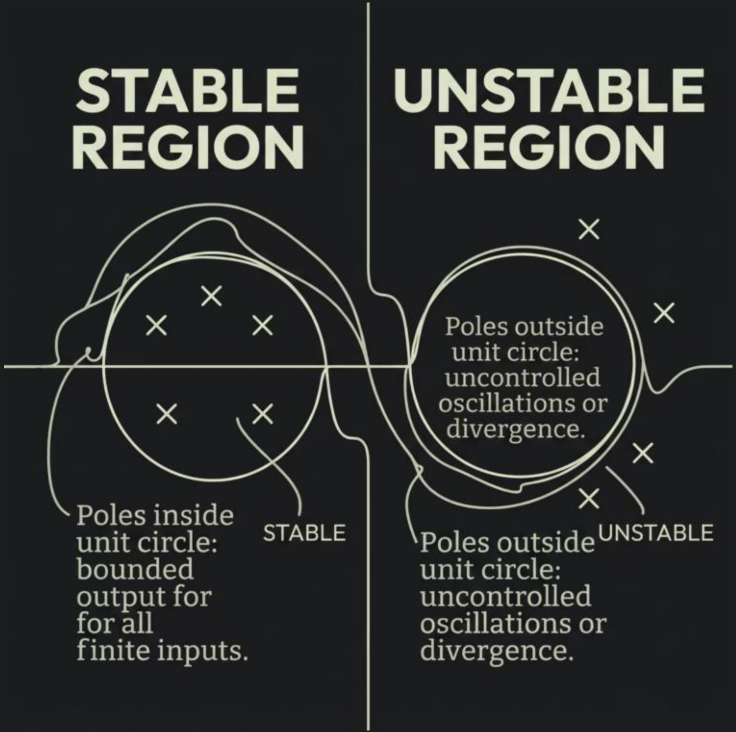
Current and past inputs u[n−k] contribute directly, just as in FIR filters.

ĞĦ FŎŌPǾMŒP

In FIR filters, all $a_k = 0$ — no feedback, no recursion.

Stability: The Tightrope Walk

The recursive nature of IIR filters introduces a critical challenge — **stability**. A filter is considered stable if its output remains bounded for any bounded input (BIBO stability).



✅ Inside Unit Circle

Poles inside → filter is stable, output decays over time.

❌ Outside Unit Circle

Poles outside → uncontrolled oscillations, output grows indefinitely.

⚠ Stability is entirely governed by the placement of poles in the z-plane — always verify before deploying an IIR design.

Phase Characteristics: A Trade-off off for Efficiency

The Core Issue

IIR filters inherently exhibit a **non-linear phase response**. This means different frequency components experience different delays — distorting the shape of the output waveform.

FIR filters, by contrast, can be designed for perfect linear phase, preserving waveform shape throughout the passband.

⚠️ ĖQŎÒŇHĦ J OÑŌê

Phase linearity is critical — e.g., telecommunications, high-fidelity audio processing, ECG signal analysis.

✅ Prefer IIR When...

Phase distortion is tolerable — e.g., speech processing, equalization, real-time embedded systems where efficiency dominates.

Common Design Approaches

IIR filter design typically begins by transforming proven **continuous-time (analog) filter prototypes** into the digital domain using methods like the bilinear transform or impulse invariance.

Butterworth

Maximally flat passband. Monotonic roll-off, no ripple anywhere.

F OÑNŘCEOÑQ

Steeper roll-off than Butterworth. Ripple in passband (Type I) or stopband (Type II).

Elliptic

Steepest possible roll-off for a given order. Ripple in both bands.

DESIGN METHOD 1

Butterworth: Smooth & Steady Steady

The Butterworth filter is prized for its **maximally flat passband** — the smoothest possible frequency response in the desired band, with zero ripple anywhere in the spectrum.

- Monotonic gain decrease from passband to stopband
- No oscillations or ripple — clean, predictable behavior
- Best choice when smoothness is prioritized over sharp cutoff
- Commonly used in audio crossover networks and sensor signal conditioning

DESIGN METHOD 2

Chebyshev: The Ripple Trade-off

Chebyshev filters accept **controlled ripple** in one band to achieve a steeper transition than Butterworth of the same order.

İ ŘŐÑHf ĨMŒŒNMŌŇ ĪÒŐŐÕÑ

Ripple is introduced in the **passband**. The stopband remains monotonic. Achieves a sharper roll-off than Butterworth for the same filter order. Ideal when passband variation is tolerable.

Type II — Stopband Ripple

Ripple is introduced in the **stopband**. The passband remains flat and monotonic. Useful when a smooth passband is required but stopband attenuation can fluctuate.

Chebyshev filters strike a deliberate balance between passband flatness and transition band steepness — a powerful middle-ground between Butterworth and Elliptic designs.

DESIGN METHOD 3

Elliptic: Maximum Steepness

Elliptic (Cauer) filters achieve the **steepest possible transition band** for any given filter order — but this performance comes with a cost: ripple in *both* the passband and the stopband.

Ì MŔÒÖ ÞÖ ĜŅŅÒŃÒÑŌŃŘ

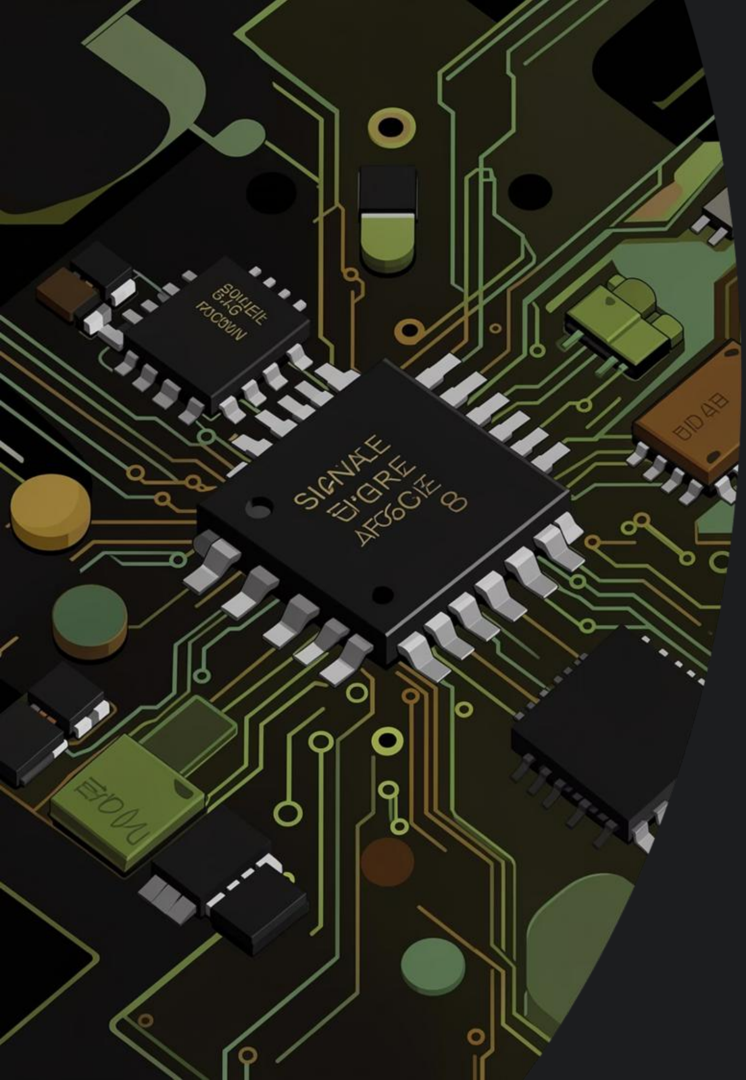
Lowest order needed for a given set of frequency specifications — minimizes computation.

Dual-Band Ripple

Equiripple in both passband and stopband — requires tolerance from the application.

Best Use Case

Applications demanding the sharpest cutoff: anti-aliasing, channel separation, RF filtering.



IIR Filters: Efficiency Meets Complexity

Computational Edge

Achieve sharp responses with fewer coefficients — ideal for real-time and embedded systems.

Stability Discipline

Recursive feedback demands careful pole placement — all poles must remain inside the unit circle.

Design Flexibility

Butterworth, Chebyshev, and Elliptic methods let engineers tailor the filter to any application requirement.

ĘǾŎMŇ Ė ŐŐŐŇMŘŎŎŒ

From audio equalization to speech processing, telecommunications, and IoT sensor filtering.

THANK YOU